

Section : Biology

Q.1 Which one of the followings is not a feature of insecticidal protein crystals formed by *Bacillus thuringiensis*: -

1. When formed they exist as inactive protoxin.
2. They are converted into an active form of toxin in the alkaline pH of the gut, which solubilize the crystals.
3. They are effective against certain insects such as lepidopterans, coleopterans and dipterans.
4. The activated toxin does not bind the surface of midgut epithelial cells which leads to pore formation, cell swelling and lysis.

Options 1. 1

2. 2
3. 3
4. 4

Q.2 The reason for the survival of the melanised moth, during the post-industrialization period in England was:

1. The melanised moth camouflage with the tree trunks that became dark due to the deposition of industrial smoke and soot so that melanised moth remain unnoticed by predators.
2. Melanised moth were able to tolerate and adapted to the high pollution level due to industrialization where as white winged moth were not able to adapt the high pollution level.
3. The lichens on the tree trunk were food only for the white winged moth and due to industrialisation lichens vanished and thus white winged moth died due to the scarcity of the food and melanised moth got survived.
4. Melanised moth were naturally selected and fittest to survive where white winged were not.

Options 1. 1

2. 2
3. 3
4. 4

Q.3 The rate of biomass production is called productivity and it is expressed in terms of:

1. $\text{g m}^{-1} \text{yr}^2$ or $(\text{kcal m}^{-1}) \text{yr}^2$
2. $\text{g m}^{-1} \text{yr}^{-1}$ or $(\text{kcal m}^{-1}) \text{yr}^{-1}$
3. $\text{g m}^{-2} \text{yr}^2$ or $(\text{kcal m}^{-2}) \text{yr}^2$
4. $\text{g m}^{-2} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Oral contraceptive pills used by females, function through:

1. Phagocytosis of sperms within uterus
2. Inhibiting ovulation and implantation
3. Suppressing sperm motility and fertilising capacity of sperm
4. Blocking gamete transport and thereby preventing conception

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Arrange the given steps involved in gel electrophoresis used for separation of DNA fragments?

- (A) Exposure to UV light
- (B) Staining with ethidium bromide
- (C) Movement of DNA fragments towards anode
- (D) Elution

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (A), (D), (C)
3. (A), (D), (B), (C)
4. (C), (B), (A), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Match List-I with List-II

List-I (Geological periods) List-II (Plant forms)

- | | |
|-------------------|---------------------|
| (A) Carboniferous | (I) Progymnosperms |
| (B) Cretaceous | (II) Monocotyledons |
| (C) Tertiary | (III) Gnetales |
| (D) Devonian | (IV) Seed ferns |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Cryopreservation technique used for the conservation of threatened animals and plants is:

1. Narrowly utilitarian approach
2. *In situ* conservation approach
3. *In vivo* approach of conservation
4. *Ex situ* conservation approach

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Amniocentesis should not be used for:

1. The diagnosis of maternal health during pregnancy
2. Testing the presence of genetic disorder in the developing embryo
3. Sex determination of the developing foetus
4. Testing any anomaly in the developing embryo

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Transgenic animals that produce useful biological products can be created by the introduction of the portion of DNA (or genes) which codes for a particular product such as human protein (α -1-antitrypsin). This protein is used to treat:

1. Pneumonia
2. Cystic fibrosis
3. Whooping cough
4. Emphysema

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Match List-I with List-II

List-I (Population interactions) List-II (Species A / Species B)

- | | |
|------------------|-------------|
| (A) Amensalism | (I) - / - |
| (B) Commensalism | (II) + / - |
| (C) Predation | (III) + / 0 |
| (D) Competition | (IV) - / 0 |

'-' sign for detrimental interaction. '+' sign for beneficial interaction. '0' sign for neutral interaction.

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.11 Arrange the given steps involved in Polymerase Chain Reaction in correct sequence?

- (A) Annealing
- (B) Extension
- (C) Denaturation
- (D) Amplification

Choose the correct answer from the options given below:

1. (C), (A), (B), (D)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (A), (C), (D), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.12 In the context of humans, which one of the following groups will have 46 chromosomes:

1. Spermatogonia and secondary spermatocytes
2. Spermatids and ovum
3. Oogonia and spermatogonia
4. Sperm and second polar body

Options 1. 1

2. 2
3. 3
4. 4

Q.13 According to species-area relationships, the relation between species richness and area for a wide variety of taxa (angiosperms, birds, bats and freshwater fishes) turns out to be a:

1. Rectangular parabola
2. Ellipse
3. Rectangular hyperbola
4. Linear hyperbola

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Which one of the following regions will have the least number of bird species?

1. Region located near the equator
2. Region located at 71°N latitude
3. Region located at 41°N latitude

4. Region located near the pole

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Match List-I with List-II

List-I (Trophic levels in an ecosystem) List-II (Examples)

- | | |
|--------------------------|--------------------|
| (A) Fourth trophic level | (I) Fishes |
| (B) Second trophic level | (II) Phytoplankton |
| (C) First trophic level | (III) Zooplankton |
| (D) Third trophic level | (IV) Man |

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.16 In eukaryotes, RNA polymerase III transcribes:

- (A) tRNA
- (B) 5.8S rRNA
- (C) 5srRNA
- (D) snRNA

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Arrange the following steps of the process of recombinant DNA technology in correct order.

- (A) Joining of foreign DNA to plasmid
- (B) Isolation and cutting of DNA at specific locations
- (C) Obtaining the foreign gene product
- (D) Insertion of rDNA or ligated DNA into the host cell/organism

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Identify the correct sequence of stages in the process of decomposition.

- (A) Humification
- (B) Fragmentation
- (C) Catabolism
- (D) Mineralisation

Choose the correct answer from the options given below:

- 1. (B), (C), (A), (D)
- 2. (A), (B), (C), (D)
- 3. (B), (A), (D), (C)
- 4. (D), (C), (B), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.19 Which one of the mentioned characters does not represent the true breeding line?

- 1. A true breeding line is one that shows the stable trait inheritance.
- 2. A true breeding line is one that shows the X-linked recessive trait.
- 3. A true breeding line is one that having undergone continuous self-pollination.
- 4. A true breeding line is one that shows stable expression for several generations.

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.20 The key tools of recombinant DNA technology are:

- (A) Ligases
- (B) Restriction enzymes
- (C) Polymerase enzymes
- (D) Suitable vectors and host organisms

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.21 For the protection from herbivore animals, the plant *Calotropis* secretes the highly poisonous compound known as -

- 1. Cardiac nucleotides
- 2. Cardiac glycoproteins
- 3. Cardiac glycosides
- 4. Cardiac globulins

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.22 Which one of the following plants will have many ovules in their gynoecium?

- 1. Paddy
- 2. Wheat
- 3. Watermelon
- 4. Mango

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.23 In organisms, the same structure developed along different directions due to adaptations to different needs. This refers to:

1. Convergent evolution
2. Divergent evolution
3. Evolution by anthropogenic action
4. Natural selection

Options 1. 1

2. 2
3. 3
4. 4

Q.24 The capacity to generate a whole plant from any cell is called:

1. Tissue culture
2. Micro-propagation
3. Parthenogenesis
4. Totipotency

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Identify the correct combination of salient features of the genetic code.

- (A) AUG has dual functions as it codes for methionine (met), and also act as initiator codon.
(B) Some amino acids are coded by more than one codon, hence the code is degenerate.
(C) UAA, UAG, UGG are stop terminator codons.
(D) The codon is read in mRNA in a continuous fashion and there are no punctuations.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.26 A nematode, *Meloidogyne incognita* infects the roots of _____ and causes a great reduction in yield.

1. Potato
2. Tomato
3. Tobacco
4. Cotton

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The similarities between sickle-cell anaemia and thalassemia are:

- (A) Both are autosome linked recessive disease
(B) Both are blood related disorder
(C) Parents of the offsprings are carriers
(D) Both are controlled by same genes

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3

Q.28 Arrange the following layers of microsporangium starting from outer to inner order–

- (A) Tapetum
- (B) Endothecium
- (C) Middle layers
- (D) Epidermis

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (B), (C), (A)
3. (B), (A), (D), (C)
4. (A), (D), (B), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Match List-I with List-II

List-I (Hormones) List-II (Functions)

- | | |
|--------------|--|
| (A) LH | (I) Necessary for maintenance of pregnancy |
| (B) GnRH | (II) Stronger uterine contractions |
| (C) Oxytocin | (III) Stimulate secretion of gonadotropins |
| (D) Relaxin | (IV) Ovulation |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Identify the incorrect feature of human genome.

1. The average gene consists of 3000 bases, but sizes vary greatly, with the largest known human gene being dystrophin at 2.4 million bases.
2. More than 8 percent of the genome codes for proteins.
3. Repeated sequences make up very large portion of the human genome.
4. Almost all (99.9 per cent) nucleotide bases are exactly the same in all people.

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The production of beta-galactosidase by the *lac* operon is started when:

1. Lactose binds to the repressor protein and inactivate it, thus RNA polymerase gets access to the promoter and then transcription proceeds where *y* gene codes for beta-galactosidase.
2. Lactose binds to the repressor protein and inactivate it, thus RNA polymerase gets access to the promoter and then transcription proceeds where *z* gene codes for beta-galactosidase.
3. Lactose binds to the repressor protein and inactivate it, thus RNA polymerase gets access to the promoter and then transcription proceeds where *a* gene codes for beta-galactosidase.
4. Lactose binds to promoter and activate it, thus RNA polymerase gets full access to the promoter and then transcription proceeds where *z* gene codes for beta-galactosidase.

Options 1. 1

2. 2
3. 3

Q.32 Which one of the following structures is not the male sex accessory duct in human?

1. Vas deferens
2. Vasa efferentia
3. Seminiferous tubules
4. Epididymis

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which of the following techniques are used for early molecular diagnosis of disease?

- (A) Urine analysis
- (B) PCR
- (C) ELISA
- (D) RDT

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C), (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.34 In India, the number of biosphere reserves and national parks are -

1. 25 biosphere reserves and 100 national parks
2. 14 biosphere reserves and 90 national parks
3. 33 biosphere reserves and 70 national parks
4. 20 biosphere reserves and 110 national parks

Options 1. 1

2. 2
3. 3
4. 4

Q.35 According to Gause's Competitive Exclusion Principle:

1. Two closely related species competing for the same resources can co-exist indefinitely and the competitively superior one will be eliminated eventually.
2. Two closely related species competing for the different resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.
3. Two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.
4. Two closely related species competing for the different resources can co-exist indefinitely and the competitively superior one will be eliminated eventually.

Options 1. 1

2. 2
3. 3
4. 4

Q.36 *Hind II* is a restriction endonuclease which always cuts DNA molecule at a particular point by recognizing:

1. Specific promoter region of ten base pairs
2. Specific sequence of four base pairs
3. Specific sequence of six base pairs
4. Specific sequence of eight base pairs

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Identify the hormone releasing IUDs -

1. Lippes loop
2. Multiload 375
3. LNG-20
4. Saheli

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Chasmogamous flowers are -

1. Dissimilar to flowers of other species with exposed anthers and stigma
2. Similar to flowers of other species with exposed anthers and stigma
3. Similar to flowers of other species with fused anthers and stigma
4. Flowers which do not open at all

Options 1. 1

2. 2

3. 3

4. 4

Q.39 Match List-I with List-II

List-I

List-II

Disorders

Features

- | | |
|-----------------------|---|
| (A) Colour Blindness | (I) 45 with 'XO' |
| (B) Haemophilia | (II) Trisomy of 21 |
| (C) Down's Syndrome | (III) Mutation of certain genes on 'X' chromosome |
| (D) Turner's Syndrome | (IV) Protein involved in clotting of blood |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.40 In humans, the fertilization takes place at -

1. Cervix
2. Isthmus region of fallopian tube
3. Infundibulum region of fallopian tube
4. Ampullary region of fallopian tube

Options 1. 1

2. 2

3. 3

4. 4

Q.41 Read the following passage carefully and answer the given questions.

During anaerobic sludge digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form biogas and can be used as source of energy as it is inflammable. The bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. You must have noticed that bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of pectinases and proteases. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by

competitively inhibiting the enzyme responsible for synthesis of cholesterol. The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water.

Cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by -

1. *Aspergillus niger*
2. *Monascus purpureus*
3. *Trichoderma polysporum*
4. *Bacillus thuringiensis*

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the following passage carefully and answer the given questions.

During anaerobic sludge digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form biogas and can be used as source of energy as it is inflammable. The bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. You must have noticed that bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of pectinases and proteases. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol. The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water.

BOD is a measure of-

1. Solid waste present in the water
2. Heavy metals present in the water
3. Inorganic matter present in the water
4. Organic matter present in the water

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the following passage carefully and answer the given questions.

During anaerobic sludge digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form biogas and can be used as source of energy as it is inflammable. The bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. You must have noticed that bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of pectinases and proteases. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol. The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water.

During anaerobic sludge digestion, bacteria produce the biogas. The composition of this biogas is -

1. Methane, hydrogen and carbon dioxide
2. Nitrogen, hydrogen sulphide and carbon dioxide
3. Methane, hydrogen sulphide and sulphur dioxide
4. Methane, hydrogen sulphide and carbon dioxide

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the following passage carefully and answer the given questions.

During anaerobic sludge digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form biogas and can be used as source of energy as it is inflammable. The bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. You must have noticed that bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of pectinases and proteases. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol. The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water.

The bottled juices are clarified by the combination of -

1. Proteases and streptokinase
2. *Saccharomyces* and *Penicillium*
3. Pectinases and proteases
4. Pectinases and *Trichoderma*

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the following passage carefully and answer the given questions.

During anaerobic sludge digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form biogas and can be used as source of energy as it is inflammable. The bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. You must have noticed that bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of pectinases and proteases. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol. The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water.

Statins, used as blood-cholesterol lowering agents are produced by -

1. *Saccharomyces cerevisiae*
2. *Trichoderma polysporum*
3. *Monascus purpureus*
4. *Staphylococci*

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the following passage carefully and answer the given questions.

Cancerous cells just continue to divide giving rise to masses of cells called tumors. Benign tumors normally remain confined to their original location and do not spread to other parts of the body and cause little damage. The malignant tumors, on the other hand are a mass of proliferating cells called neoplastic or tumor cells. Transformation of normal cells into cancerous neoplastic cells may be induced by physical, chemical or biological agents. Ionising radiations like X-rays and gamma rays and non-ionizing radiations like UV cause cancer. Cancer-causing viruses are called oncogenic viruses which have viral oncogenes. The substances to which an immune response is produced are called allergens. The antibodies produced to these are of IgE type. Common examples of allergens are mites in dust, pollens, animal dander, etc. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

The tumors that remain confined to their original location and do not spread to other parts of the body are:

1. Malignant tumors
2. Benign tumors

- 3. Oncogenes
- 4. Carcinogens

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.47 Read the following passage carefully and answer the given questions.
Cancerous cells just continue to divide giving rise to masses of cells called tumors. Benign tumors normally remain confined to their original location and do not spread to other parts of the body and cause little damage. The malignant tumors, on the other hand are a mass of proliferating cells called neoplastic or tumor cells. Transformation of normal cells into cancerous neoplastic cells may be induced by physical, chemical or biological agents. Ionising radiations like X-rays and gamma rays and non-ionizing radiations like UV cause cancer. Cancer-causing viruses are called oncogenic viruses which have viral oncogenes. The substances to which an immune response is produced are called allergens. The antibodies produced to these are of IgE type. Common examples of allergens are mites in dust, pollens, animal dander, etc. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

Which one of the following antibodies are produced in response to the allergens?

- 1. IgA
- 2. IgG
- 3. IgE
- 4. IgM

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.48 Read the following passage carefully and answer the given questions.
Cancerous cells just continue to divide giving rise to masses of cells called tumors. Benign tumors normally remain confined to their original location and do not spread to other parts of the body and cause little damage. The malignant tumors, on the other hand are a mass of proliferating cells called neoplastic or tumor cells. Transformation of normal cells into cancerous neoplastic cells may be induced by physical, chemical or biological agents. Ionising radiations like X-rays and gamma rays and non-ionizing radiations like UV cause cancer. Cancer-causing viruses are called oncogenic viruses which have viral oncogenes. The substances to which an immune response is produced are called allergens. The antibodies produced to these are of IgE type. Common examples of allergens are mites in dust, pollens, animal dander, etc. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

Cancer cells divide without any control due to:

- 1. Presence of contact inhibition between cells
- 2. No contact between the cells
- 3. Loss of DNA transcription
- 4. Loss of contact inhibition between cells

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.49 Read the following passage carefully and answer the given questions.
Cancerous cells just continue to divide giving rise to masses of cells called tumors. Benign tumors normally remain confined to their original location and do not spread to other parts of the body and cause little damage. The malignant tumors, on the other hand are a mass of proliferating cells called neoplastic or tumor cells. Transformation of normal cells into cancerous

neoplastic cells may be induced by physical, chemical or biological agents. Ionising radiations like X-rays and gamma rays and non-ionizing radiations like UV cause cancer. Cancer-causing viruses are called oncogenic viruses which have viral oncogenes. The substances to which an immune response is produced are called allergens. The antibodies produced to these are of IgE type. Common examples of allergens are mites in dust, pollens, animal dander, etc. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

Which one of the followings cannot cause cancer:

1. Chemicals present in tobacco
2. Oncogenic viruses
3. Physical contact with the cancer patient
4. Ionizing radiation like X-rays and gamma rays

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the following passage carefully and answer the given questions.

Cancerous cells just continue to divide giving rise to masses of cells called tumors. Benign tumors normally remain confined to their original location and do not spread to other parts of the body and cause little damage. The malignant tumors, on the other hand are a mass of proliferating cells called neoplastic or tumor cells. Transformation of normal cells into cancerous neoplastic cells may be induced by physical, chemical or biological agents. Ionising radiations like X-rays and gamma rays and non-ionizing radiations like UV cause cancer. Cancer-causing viruses are called oncogenic viruses which have viral oncogenes. The substances to which an immune response is produced are called allergens. The antibodies produced to these are of IgE type. Common examples of allergens are mites in dust, pollens, animal dander, etc. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

The primary lymphoid organs are:

1. Appendix and spleen
2. Tonsils and Peyer's patches of small intestine
3. Bone marrow and lymph nodes
4. Thymus and bone marrow.

Options 1. 1

2. 2
3. 3
4. 4